DARWIN AND EVOLUTION

Module map

- Module 12: Darwin and Evolution
- Connected lab: lab-13_darwin-evolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 12

Darwin and Evolution

1. Historical explanation for adaptation
2. Variation and heritability
3. Natural selection mechanism
4. Evidence for common ancestry
5. Fitness as reproductive success

Lab evidence

lab-13_darwin-evolution.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

DARWIN AND EVOLUTION

Learning objectives

- Explain natural selection using variation, heritability, and reproductive success.
- Define fitness in evolutionary terms.
- Distinguish individual change from population evolution.
- Use multiple evidence types to support common ancestry.
- Apply natural selection to a concrete trait example.

OBJECTIVE 1
Explain natural selection using variation, heritability, and reproductive success.

OBJECTIVE 2
Define fitness in evolutionary terms.

OBJECTIVE 3
Distinguish individual change from population evolution.

OBJECTIVE 4
Use multiple evidence types to support common ancestry.

OBJECTIVE 5
Apply natural selection to a concrete trait example.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

DARWIN AND EVOLUTION

Topic sequence

- Historical explanation for adaptation: Descent with modification links inherited similarities and differences to shared ancestors. Evolution has no goal or ladder of progress.
- Variation and heritability: Natural selection sorts existing heritable variation through unequal reproductive success; need does not create a useful mutation.
- Natural selection mechanism: Fitness is reproductive success relative to others in the same population and environment. Strength or survival alone is insufficient.
- Evidence for common ancestry: Fossils, homologous anatomy, biogeography, and DNA support common ancestry. Similar function alone can reflect convergence.
- Fitness as reproductive success: Individuals can acclimate; populations evolve across generations. Test selection with evidence of inheritance and reproductive differences.

01

Historical explanation for adaptation

Descent with modification links inherited similarities and differences to shared ancestors. Evolution has no goal or ladder of progress.

02

Variation and heritability

Natural selection sorts existing heritable variation through unequal reproductive success; need does not create a useful mutation.

03

Natural selection mechanism

Fitness is reproductive success relative to others in the same population and environment. Strength or survival alone is insufficient.

04

Evidence for common ancestry

Fossils, homologous anatomy, biogeography, and DNA support common ancestry. Similar function alone can reflect convergence.

05

Fitness as reproductive success

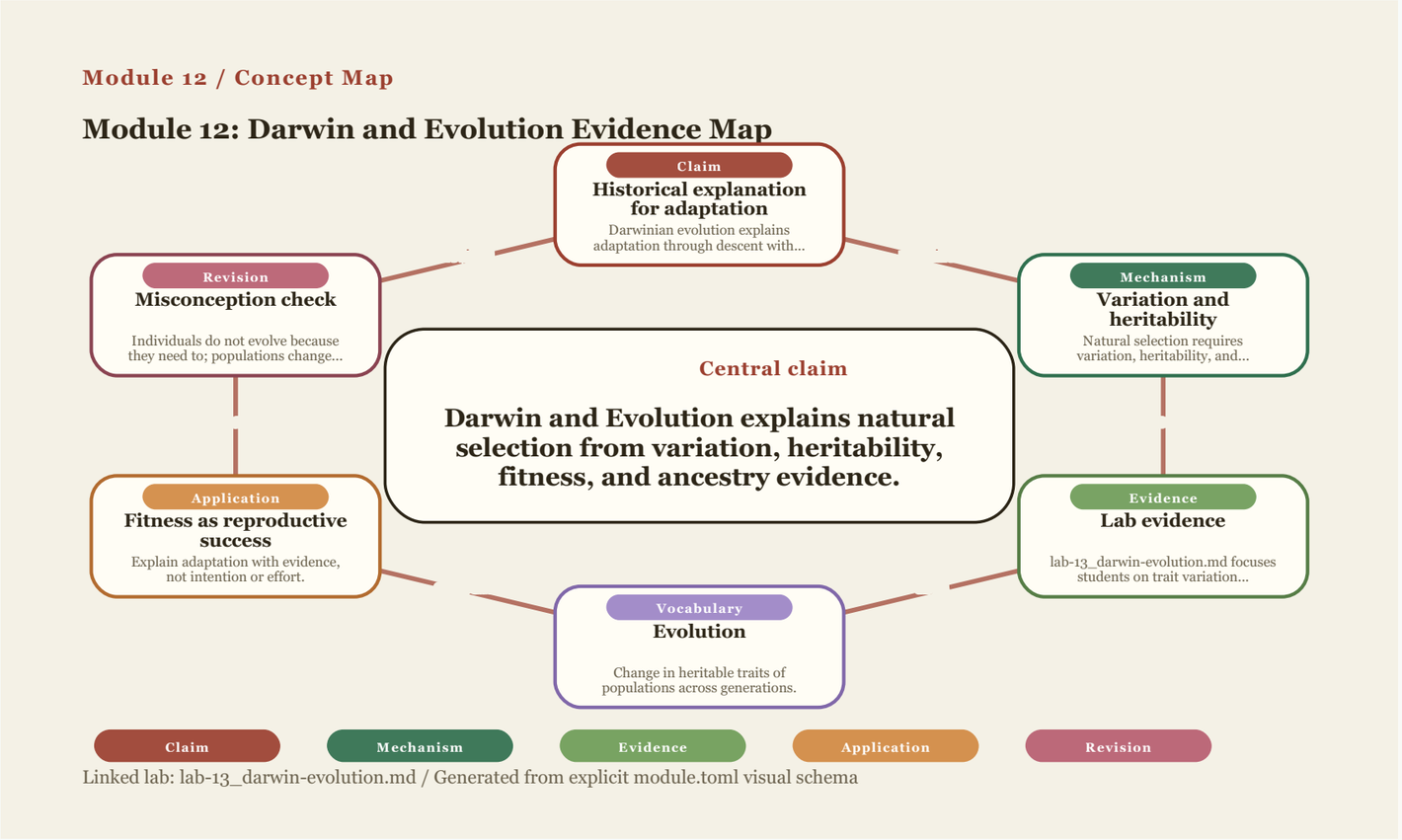
Individuals can acclimate; populations evolve across generations. Test selection with evidence of inheritance and reproductive differences.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

DARWIN AND EVOLUTION

Module 12: Darwin and Evolution Evidence Map

- Module focus: Darwin and Evolution
- Central claim: Darwin and Evolution explains natural selection from variation, heritability, fitness, and ancestry evidence.
- Key nodes to connect: Historical explanation for adaptation, Variation and heritability, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-13_darwin-evolution.md supplies an evidence surface for the map.

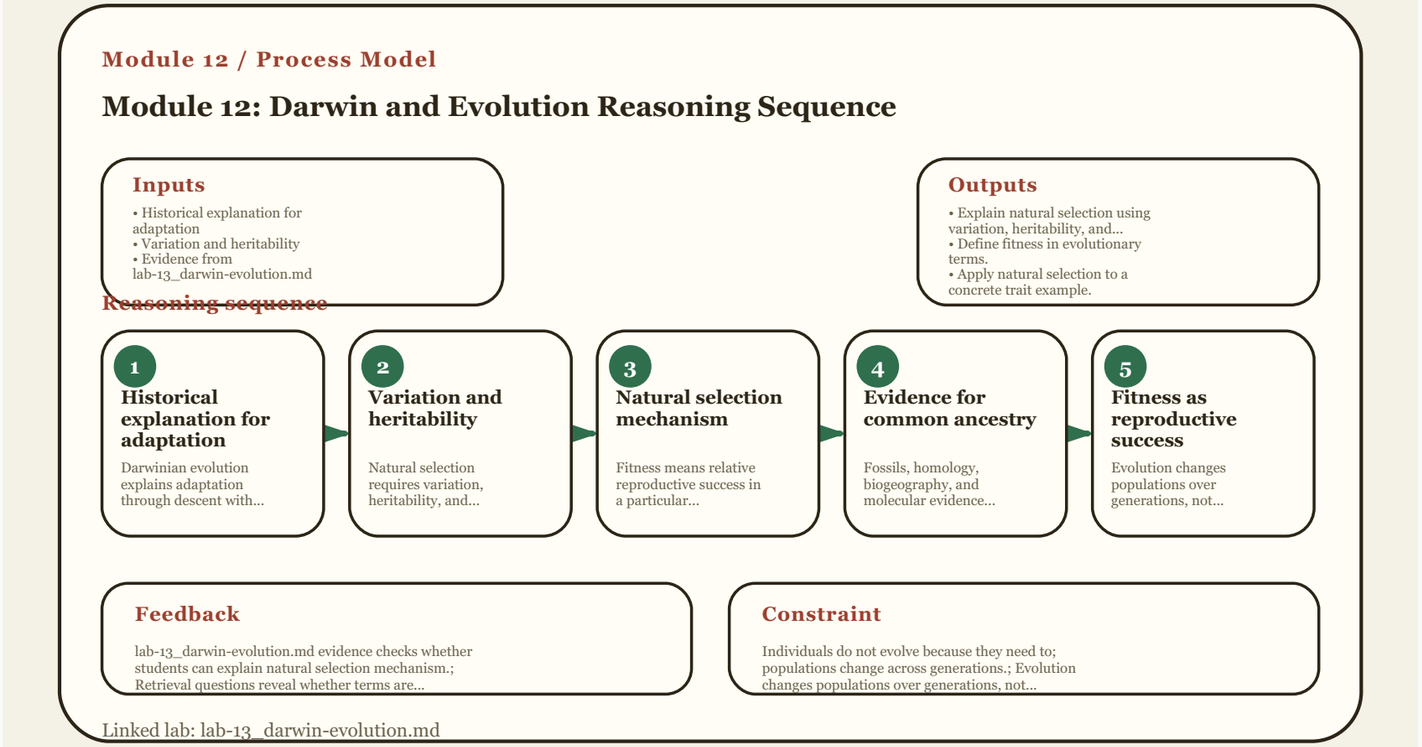


Teaching note: Ask students to explain one edge in their own words before moving on.

DARWIN AND EVOLUTION

Module 12: Darwin and Evolution Reasoning Sequence

- Module focus: Darwin and Evolution
- Trace the model: Historical explanation for adaptation -> Variation and heritability -> Natural selection mechanism -> Evidence for common ancestry
- Inputs become outputs: Historical explanation for adaptation, Variation and heritability -> Explain natural selection using variation, heritability, and reproductive success., Define fitness in evolutionary terms.
- Feedback check: lab-13_darwin-evolution.md evidence checks whether students can explain natural selection mechanism.
- Lab connection: lab-13_darwin-evolution.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

DARWIN AND EVOLUTION

Terms as evidence handles

- Evolution: Change in heritable traits of populations across generations.
- Natural selection: Nonrandom differences in survival or reproduction tied to heritable traits.
- Adaptation: Heritable trait shaped by selection in a context.
- Fitness: Relative reproductive success.
- Common ancestry: Shared origin of lineages through descent.
- Homology: Similarity due to shared ancestry.

EVOLUTION
Change in heritable traits of populations across generations.

NATURAL SELECTION
Nonrandom differences in survival or reproduction tied to heritable traits.

ADAPTATION
Heritable trait shaped by selection in a context.

FITNESS
Relative reproductive success.

COMMON ANCESTRY
Shared origin of lineages through descent.

HOMOLOGY
Similarity due to shared ancestry.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

DARWIN AND EVOLUTION

Lab connection

- Lab file: lab-13_darwin-evolution.md
- Lab evidence should test or illustrate: Natural selection mechanism
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Historical explanation for adaptation

EVIDENCE
lab-13_darwin-evolution.md

CLAIM
Explain natural selection using variation, heritability, and reproductive success.

Teaching note: Make the lab connection explicit before students encounter the activity.

DARWIN AND EVOLUTION

Contrast check

- Do not stop at: Descent with modification links inherited similarities and differences to shared ancestors. Evolution has no goal or ladder of progress.
- Stronger explanation: Individuals can acclimate; populations evolve across generations. Test selection with evidence of inheritance and reproductive differences.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Descent with modification links inherited similarities and differences to shared ancestors. Evolution has no goal or ladder of progress.

DEEPER

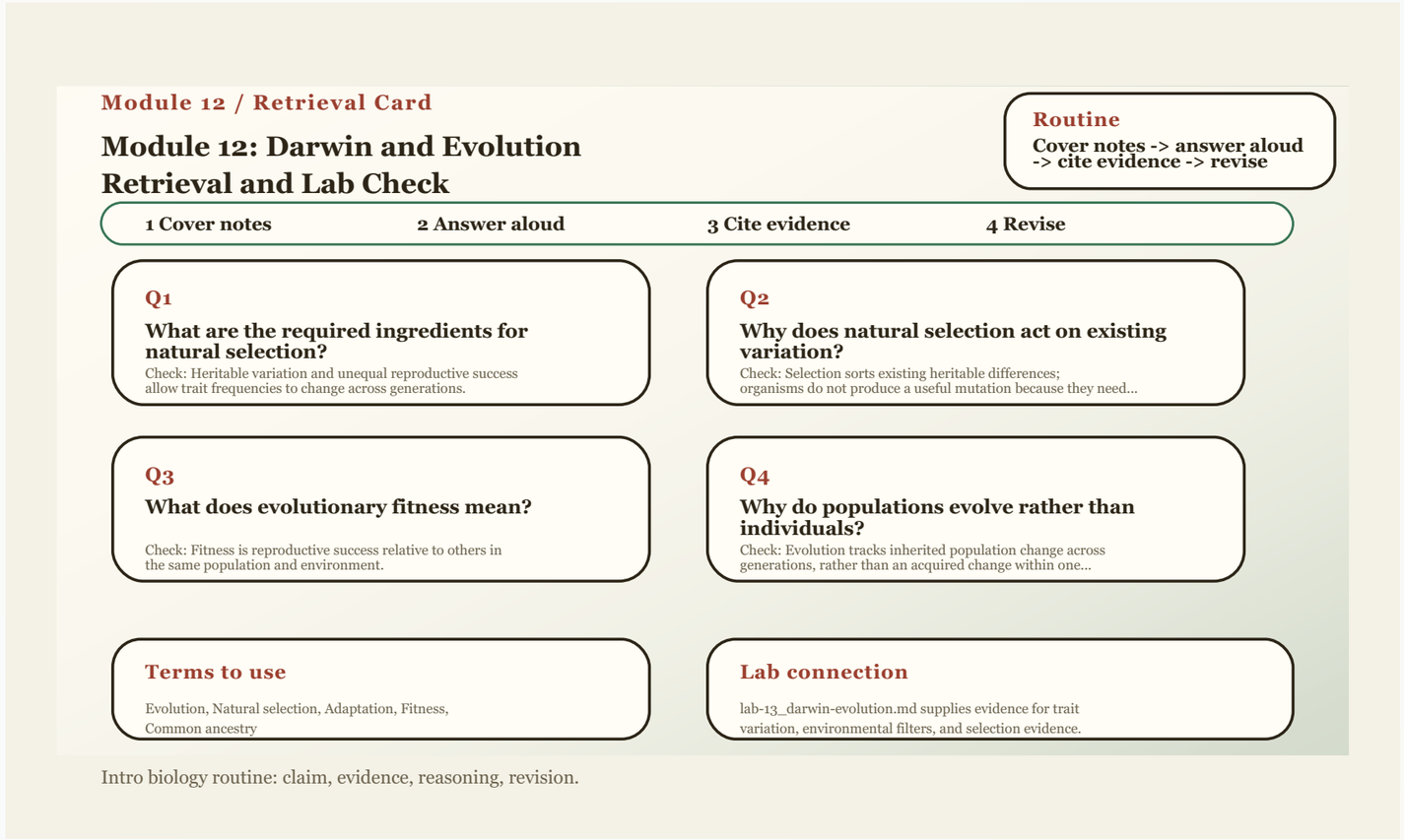
Individuals can acclimate; populations evolve across generations. Test selection with evidence of inheritance and reproductive differences.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

DARWIN AND EVOLUTION

Module 12: Darwin and Evolution Retrieval and Lab Check

- Module focus: Darwin and Evolution
- Retrieval prompt: What are the required ingredients for natural selection?
- Answer check: Heritable variation and unequal reproductive success allow trait frequencies to change across generations.
- Required terms: Evolution, Natural selection, Adaptation
- Lab connection: lab-13_darwin-evolution.md; lab-13_darwin-evolution.md supplies evidence for trait variation, environmental filters, and selection evidence.



Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

DARWIN AND EVOLUTION

Practice quiz bridge

- 1. Natural selection requires all of the following EXCEPT:
- 2. In evolutionary terms, an organism's "fitness" means its:
- 3. The forelimb bones of humans, whales, and bats share the same underlying arrangement. This is evidence of:
- 4. "An individual giraffe stretched its neck during its life and so it evolved a longer neck." Why is this wrong?

QUESTION 1

Key: C

Natural selection requires all of the following EXCEPT:

QUESTION 2

Key: A

In evolutionary terms, an organism's "fitness" means its:

QUESTION 3

Key: D

The forelimb bones of humans, whales, and bats share the same underlying arrangement. This is evidence of:

QUESTION 4

Key: B

"An individual giraffe stretched its neck during its life and so it evolved a longer neck." Why is this wrong?

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

DARWIN AND EVOLUTION

Synthesis and exit ticket

- Exit claim: explain historical explanation for adaptation using evidence from lab-13_darwin-evolution.md.
- Must include: Evolution, Natural selection, Adaptation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Historical explanation for adaptation

EVIDENCE
lab-13_darwin-evolution.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.